

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that: when a DNS request is sent to the DNS server the port is unreachable.

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: udp port 53 unreachable

The port noted in the error message is used for: DNS services.

The most likely issue is: from the DNS service. It was unavailable or not listening on UDP port 53. Firewall configuration blocking access to UDP port 53 may be a possible issue too.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred: The incident happened in the afternoon around 1:24pm.

Explain how the IT team became aware of the incident: The IT department became aware of the incident after several customers reported that they could not access the website and they received a 'destination port is unreachable' error message. The analyst attempted to access the website and experience the same error message;

Explain the actions taken by the IT department to investigate the incident: The analyst first confirmed the issue by first accessing the website where the same error message was experienced. The analyst then troubleshoots the issue using the tcpdump network protocol analyzer to analyze the packets.

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.): After analyzing the packets, it showed that requests were sent from the client to the DNS server using the UDP. In response the ICMP sent an error message 'udp port 53 unreachable' indicating that the requests could not reach the udp port 53.

Note a likely cause of the incident: The clear cause for now is that the DNS service is unavailable on the udp port 53. Another cause maybe that some firewall configuration is blocking access to the udp port 53